Nishal Stanislaus Silva, *PhD*

Senior Machine Learning Engineer | Deep Learning, Time-Series, Scalable ML Systems | Applied ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

nishal.silva@hotmail.com | +1 613 200 2030 | Toronto, ON | nishal.xyz | linkedin.com/in/nishal-silva

---

19th October 2026

Re: Senior Machine Learning Engineer – TD (Layer 6)

Dear Hiring Manager,

I am applying for the Senior Machine Learning Engineer role at Layer 6. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia, and I build end-to-end ML systems that run in production under hard constraints. Layer 6 is attractive to me because it ships ML into a large regulated business, works on real transaction, text and document data, and still publishes and collaborates with academics. That mix of research and shipped systems is how I have worked since my PhD.

Three parts of your posting match my work directly. For production ML, I owned full pipelines for streaming multi-dimensional data on the EU-funded MUSMET project, from data acquisition and feature engineering to training, deployment and monitoring. Our published system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline, with F1 0.76 (Journal of the Audio Engineering Society, 2026). For data-intensive and financial work, I built real-time ETL pipelines for high-frequency IoT time-series across distributed sites at MAS Holdings, and designed ML and statistical pipelines for financial forensics and compliance risk detection at Forestpin. For deep learning and research, I work in PyTorch, TensorFlow and JAX, have trained generative models on GPUs, and have 10+ peer-reviewed publications.

I should be direct about the gaps. I have not worked with LangGraph or built agentic systems in production, and my banking experience is limited to the Forestpin consulting work rather than a bank's data platform. My production code is mostly Python and C++, not Java. My Gen AI work covers LLM and RAG tooling and generative models for synthetic data, not large-scale customer-facing systems. I write tested, documented code with CI/CD, and I expect to be productive quickly on your stack.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, available immediately, and comfortable with the onsite Toronto arrangement. I would welcome the opportunity to discuss how my experience can support the ML engineering team at TD (Layer 6). My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD